

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 7: Linear Phase Filters

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1 Linear Phase Filters

Example (Delay System).

$$h[n] = \delta[n - M] \quad (\text{delay system})$$

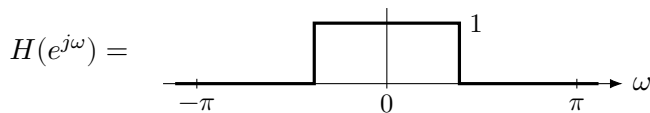
$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n - M] e^{-j\omega n} = e^{-j\omega M}.$$

$$|H(e^{j\omega})| = 1, \quad \theta(\omega) = -\omega M.$$

Here $\theta(\omega)$ is a **linear function of ω** (slope $-M$): phase grows in proportion to frequency, so **linear phase = constant delay**.

1.1 Why Linear Phase? Causal Approximation of Ideal Filters

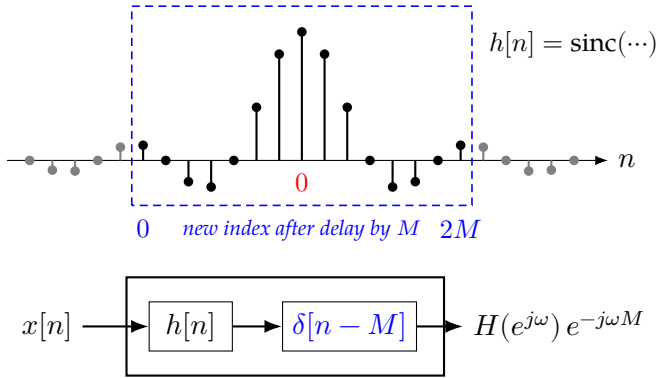
Linear phase is useful for **causal** systems. The ideal lowpass filter



has the non-causal, infinitely long impulse response

$$h[n] = \text{sinc}(\dots)$$

(symmetric about $n = 0$). *Truncate* it to a finite window $-M \leq n \leq M$ and **delay** it by M samples, so that the window starts at $n = 0$:



2 Generalized Linear Phase (GLP)

Def.

A system has **generalized linear phase (GLP)** if

$$H(e^{j\omega}) = \underbrace{R(e^{j\omega})}_{\text{real function}} e^{j(k_1 + \omega k_2)},$$

where $R(e^{j\omega})$ is real-valued (i.e., has phase 0 or π), and k_1, k_2 are constants.

GLP filters: **linear phase in the passband** (the sign flips of R only add jumps of π).

Property.

A **sufficient** condition for GLP for a real-valued, causal, length- N FIR filter is that

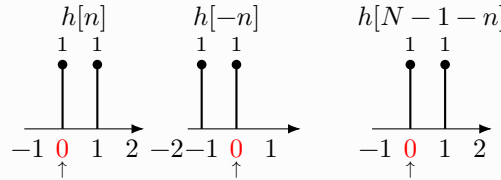
$$h[n] = \pm h[N - 1 - n] \quad \text{for } \forall n,$$

with $h[n] = 0 \quad \forall n < 0, n \geq N$.

If $h[0] \neq 0$ and $h[N - 1] \neq 0$, then the condition is also **necessary**.

Examples.

- $h[n] = \{1, 1\}$, $h[-n] = \{1, 1\}$, $h[N-1-n] = \{1, 1\}$ ✓



↑ marks $n = 0$ on each axis.

- $h[n] = \{1, -1\}$, $h[N-1-n] = \{-1, 1\}$ ✓
- $h[n] = \{1, -1, 1, -1\}$, $h[N-1-n] = \{-1, 1, -1, 1\}$ ✓ ($N = 4$, Type 4 antisymmetric)
- $h[n] = \{-1, 2, 1\}$, $h[0] = -h[2]$ but $h[1] = -h[1]$ fails ✗

Odd-length filters with antisymmetry cannot satisfy GLP unless the middle sample is 0.

with $h[n] = 0$ for $n \geq N$ and $n < 0$.

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2.1 Four Types of FIR GLP Filters

- Even symmetric or antisymmetric (+ or - in $h[n] = \pm h[N-1-n]$),
- N odd or even.

Notation: this handout vs. Mitra.

In this handout, N denotes the FIR filter **length**: the samples are $h[0], \dots, h[N-1]$, and the filter order is $N-1$. In Mitra Ch. 7, N denotes the filter **order**: the samples are $h[0], \dots, h[N]$, and the length is $N+1$. Hence

$$N_{\text{handout}} = N_{\text{Mitra}} + 1.$$

The Type 1–4 definitions are the same; only the meaning of the symbol N differs. For example, Mitra's Type 3 has even order N , hence odd length $N+1$, corresponding to §2.1.2 below.

2.1.1 N odd, symmetric and its proof (Mitra Type 1)

Assume $N = 2M + 1 \iff M = N - M - 1$, and $h[n] = h[N-1-n]$ for all n . Then

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=0}^{N-1} h[n] e^{-j\omega n} \\ &= \sum_{n=0}^{M-1} h[n] e^{-j\omega n} + \underbrace{h[M] e^{-j\omega M}}_{\text{middle point}} + \sum_{n=M+1}^{N-1} h[n] e^{-j\omega n} \\ &= e^{-j\omega M} \left(h[M] + \sum_{n=0}^{M-1} h[n] e^{-j\omega(n-M)} + \boxed{\sum_{n=M+1}^{N-1} h[n] e^{-j\omega(n-M)}} \right). \end{aligned}$$

In the **last sum**, let $n = N-1-k \iff k = N-1-n$:

$$\sum_{k=M-1}^0 h[N-1-k] e^{-j\omega(N-1-k-M)} = \sum_{k=0}^{M-1} h[N-1-k] e^{-j\omega(M-k)}.$$

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Renaming $\mathbf{k} \rightarrow \mathbf{n}$ and using $h[N - 1 - n] = h[n]$:

$$\begin{aligned}
 H(e^{j\omega}) &= e^{-j\omega M} \left[h[M] + \sum_{n=0}^{M-1} \left(\mathbf{h}[\mathbf{n}] e^{-j\omega(n-M)} \right. \right. \\
 &\quad \left. \left. + \underbrace{h[N - 1 - n]}_{\mathbf{h}[\mathbf{n}]} e^{+j\omega(\mathbf{n}-\mathbf{M})} \right) \right] \\
 &= \underbrace{e^{-j\omega M}}_{\text{linear phase}} \underbrace{\left[h[M] + \bullet 2 \sum_{n=0}^{M-1} h[n] \cos(\omega[n - M]) \right]}_{\text{real function}}. \quad \blacksquare \quad (\text{Mittra 7.53})
 \end{aligned}$$

● **Remark**

(addendum to orig lecture note).

How to see $e^{-j\omega(n-M)} + e^{+j\omega(n-M)} = 2 \cos(\omega(n-M))$: this follows directly from **Euler's formula**. Let

$$\theta = \omega(n-M).$$

Then consider

$$e^{-j\theta} + e^{j\theta}.$$

From Euler's formula,

$$e^{j\theta} = \cos \theta + j \sin \theta,$$

and

$$e^{-j\theta} = \cos(-\theta) + j \sin(-\theta) = \cos \theta - j \sin \theta.$$

Therefore,

$$\begin{aligned} e^{-j\theta} + e^{j\theta} &= (\cos \theta - j \sin \theta) + (\cos \theta + j \sin \theta) \\ &= 2 \cos \theta. \end{aligned}$$

Substituting $\theta = \omega(n-M)$, we obtain

$$\boxed{e^{-j\omega(n-M)} + e^{j\omega(n-M)} = 2 \cos(\omega(n-M))}.$$

The intuition is that the two complex exponentials are complex conjugates:

$$\underbrace{\cos \theta - j \sin \theta}_{e^{-j\theta}} + \underbrace{\cos \theta + j \sin \theta}_{e^{j\theta}}.$$

Their imaginary parts cancel,

$$-j \sin \theta + j \sin \theta = 0,$$

while their real parts add:

$$\cos \theta + \cos \theta = 2 \cos \theta.$$

2.1.2 N odd, antisymmetric (odd symmetry) and its proof (Mitra Type 3)

Assume $N = 2M + 1$ is odd (so $M = \frac{N-1}{2} = N - M - 1$), $h[n] \in \mathbb{R}$ for all n , and $h[n] = -h[N - 1 - n]$ for all n (Type 3).

At the center tap, $h[M] = -h[N - 1 - M] = -h[M]$, hence $h[M] = 0$. Then

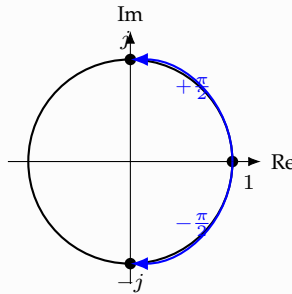
$$H(e^{j\omega}) = 2 e^{-j(\frac{\pi}{2} + \omega M)} \sum_{n=0}^{M-1} h[n] \sin(\omega[n - M]) \quad (\text{Recall: } \bullet e^{\pm j\frac{\pi}{2}} = \pm j). \quad (\text{Mitra 7.57})$$

● Remark

(addendum to orig lecture note).

Unit-circle picture for $e^{\pm j\pi/2} = \pm j$: each $e^{j\theta}$ is the point on the **unit circle** at angle θ counterclockwise from the positive real axis,

$$e^{j\theta} = \cos \theta + j \sin \theta.$$



At $\theta = +\frac{\pi}{2}$: $\cos(\frac{\pi}{2}) = 0$, $\sin(\frac{\pi}{2}) = 1$, so $e^{+j\pi/2} = j$. At $\theta = -\frac{\pi}{2}$: $e^{-j\pi/2} = -j$. Thus the two Recall values are quarter-turns from 1: **up** to $+j$, **down** to $-j$. In the formula above,

$$e^{-j(\frac{\pi}{2} + \omega M)} = \underbrace{e^{-j\pi/2}}_{\text{fixed quarter-turn} = -j} \underbrace{e^{-j\omega M}}_{\text{linear-phase factor varies with } \omega}.$$

Antisymmetry turns paired exponentials into sines and picks up this extra $\pm j$ factor—the “initial” phase offset $\pm \frac{\pi}{2}$ in the Note below—whereas symmetry ($h[n] = h[N - 1 - n]$) leaves a real cosine sum (phase 0 or π).

2.1.3 N even, antisymmetric (odd symmetry) and its proof (Mitra Type 4; from HW4)

Assume $N = 2M$ is even, $h[n] \in \mathbb{R}$ for all n , and $h[n] = -h[N - 1 - n]$ for all n (Type 4). There is no center tap at $n = \frac{N-1}{2} = M - \frac{1}{2}$; the impulse response is antisymmetric about the midpoint between samples $M - 1$ and M . Then

$$H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n] e^{-j\omega n} = \sum_{n=0}^{M-1} h[n] e^{-j\omega n} + \sum_{n=M}^{N-1} h[n] e^{-j\omega n}.$$

In the **last sum**, let $n = N - 1 - k$ (so $k = N - 1 - n$):

$$\sum_{k=0}^{M-1} h[N - 1 - k] e^{-j\omega(N-1-k)}.$$

5 Renaming **k** $\rightarrow$ **n** and using $h[N - 1 - n] = -h[n]$:

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=0}^{M-1} h[n] \left(e^{-j\omega n} - e^{-j\omega(N-1-n)} \right) \\ &= e^{-j\omega(M-\frac{1}{2})} \sum_{n=0}^{M-1} h[n] \left(e^{-j\omega(n-M+\frac{1}{2})} - e^{+j\omega(n-M+\frac{1}{2})} \right) \\ &= \underbrace{2 e^{-j(\frac{\pi}{2} + \omega(M-\frac{1}{2}))}}_{\text{linear phase}} \underbrace{\sum_{n=0}^{M-1} h[n] \sin(\omega[n - M + \frac{1}{2}])}_{\text{real function}}. \quad \blacksquare \quad (\text{Mitra 7.58}) \end{aligned}$$

The group delay is $\frac{N-1}{2} = M - \frac{1}{2}$ samples (a half-integer), as in the Type 1–4 table below; compare with §2.1.2 (Type 3), where odd N gives delay M and $\sin(\omega[n - M])$ instead of $\sin(\omega[n - M + \frac{1}{2}])$.

2.1.4 N even, symmetric and its proof (Mitra Type 2)

Assume $N = 2M$ is even, $h[n] \in \mathbb{R}$ for all n , and $h[n] = h[N - 1 - n]$ for all n (Type 2). The axis of symmetry is at $n = \frac{N-1}{2} = M - \frac{1}{2}$ (between samples $M - 1$

and M ; there is no center tap). Then

$$\begin{aligned}
 H(e^{j\omega}) &= \sum_{n=0}^{N-1} h[n] e^{-j\omega n} \\
 &= \sum_{n=0}^{M-1} h[n] e^{-j\omega n} + \sum_{n=M}^{N-1} h[n] e^{-j\omega n} \\
 &= e^{-j\omega(M-\frac{1}{2})} \left(\sum_{n=0}^{M-1} h[n] e^{-j\omega(n-M+\frac{1}{2})} + \sum_{n=M}^{N-1} h[n] e^{-j\omega(n-M+\frac{1}{2})} \right).
 \end{aligned}$$

In the **last sum**, let $n = N - 1 - k \iff k = N - 1 - n$:

$$\sum_{k=M-1}^0 h[N-1-k] e^{-j\omega(N-1-k-M)} = \sum_{k=0}^{M-1} h[N-1-k] e^{-j\omega(M-1-k)}.$$

6 Renaming $\mathbf{k} \rightarrow \mathbf{n}$ and using $h[N-1-n] = h[n]$:

$$\begin{aligned}
 H(e^{j\omega}) &= e^{-j\omega(M-\frac{1}{2})} \left[\sum_{n=0}^{M-1} \left(\mathbf{h}[\mathbf{n}] e^{-j\omega(n-M+\frac{1}{2})} \right. \right. \\
 &\quad \left. \left. + \underbrace{h[N-1-n]}_{\mathbf{h}[\mathbf{n}]} e^{+j\omega(n-M+\frac{1}{2})} \right) \right] \\
 &= \underbrace{e^{-j\omega(M-\frac{1}{2})}}_{\text{linear phase}} \left[\underbrace{\bullet 2 \sum_{n=0}^{M-1} h[n] \cos(\omega[n-M+\frac{1}{2}])}_{\text{real function}} \right]. \quad \blacksquare \quad (\text{Mitra 7.56})
 \end{aligned}$$

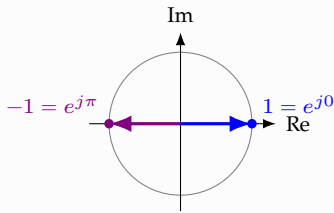
The group delay is $\frac{N-1}{2} = M - \frac{1}{2}$ samples. At $\omega = \pi$, each $\cos(\omega[n-M+\frac{1}{2}])$ is zero, so $H(e^{j\pi}) = 0$ (Type 2 cannot be a strict highpass); compare §2.1.1 (Type 1, (Mitra 7.53)), where length- N is odd and $\cos(\omega[n-M])$.

Note.

Recall from the beginning: $e^{-j\omega M}$ corresponds to a delay of M samples. GLP FIR filters are associated with a **delay of M samples** (larger $M \Rightarrow$ longer delay).

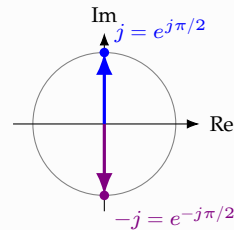
- Symmetric (+) $\Rightarrow$ “initial” phase of 0 or π .
- Antisymmetric (–) $\Rightarrow$ “initial” phase of $\pm \frac{\pi}{2} \Rightarrow$ good for: differentiation; Hilbert transform.

Symmetric (+): real axis



initial phase = 0 or π

Antisymmetric (–): imaginary axis



initial phase = $+\frac{\pi}{2}$ or $-\frac{\pi}{2}$

Remark

(addendum to orig lecture note).

Type	Symmetry	N (len.)	Forced zeros of $H(e^{j\omega})$
1	$h[n] = h[N-1-n]$	odd	none
2	$h[n] = h[N-1-n]$	even	$\omega = \pi$
3	$h[n] = -h[N-1-n]$ (odd symmetry)	odd	$\omega = 0$ and $\omega = \pi$
4	$h[n] = -h[N-1-n]$ (odd symmetry)	even	$\omega = 0$

- Type 2 cannot realize a highpass filter.
- Types 3/4 are antisymmetric (odd symmetry), have phase offset $\pm \frac{\pi}{2}$, and satisfy $H(e^{j0}) = 0$; they suit differentiators and Hilbert transformers.

5 3 Differentiators and Hilbert Transformers

3.1 Differentiator

$$\frac{dx(t)}{dt} \leftrightarrow j\Omega X(j\Omega) \implies H(e^{j\omega}) \approx j\omega$$

(periodization and approximated appropriately).

$$H_{\text{DIF}}(e^{j\omega}) = j\omega, \quad 0 \leq |\omega| \leq \pi. \quad (\text{Mittra 7.100; 10.26})$$

GLP check (§2). With $R(e^{j\omega}) = \omega$ (real; phase 0 or π) and $k_1 = \frac{\pi}{2}$, $k_2 = 0$,

$$\begin{aligned}
 H_{\text{DIF}}(e^{j\omega}) &= R(e^{j\omega}) e^{j(k_1 + \omega k_2)} \\
 &= \omega e^{j\pi/2} \quad (\text{Recall: Euler } e^{j\theta} = \cos \theta + j \sin \theta, \theta = \frac{\pi}{2} \Rightarrow e^{j\pi/2} = j) \\
 &= j\omega, \quad \omega \in [-\pi, \pi].
 \end{aligned}$$

The fixed factor $e^{j\pi/2}$ is the same *initial phase* $\frac{\pi}{2}$ as in antisymmetric Types 3/4 (§2); here $k_2 = 0$, so there is no extra linear phase term $-\omega M$ beyond that offset.

3.2 Hilbert Transformer

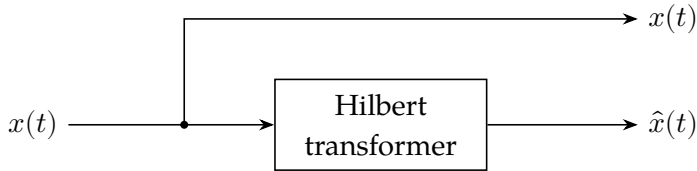
The ideal Hilbert transformer, also called a 90-degree phase shifter,¹ is characterized by a frequency response

$$H(j\Omega) = \begin{cases} -j, & \Omega \geq 0 \\ +j, & \Omega < 0 \end{cases}$$

(Here Ω is the continuous-time angular frequency, usually in rad/s.) **DTFT (ideal Hilbert transformer).** Frequency response $H(e^{j\omega})$:

$$H(e^{j\omega}) \approx \begin{cases} -j, & 0 \leq \omega < \pi \\ +j, & -\pi \leq \omega < 0 \end{cases} \quad (\text{Mitra 10.24})$$

(Here ω is the discrete-time normalized angular frequency, in rad/sample.)



Generation of an analytic signal using a Hilbert transformer. (Mitra Figure 1.3, Sec. 1.2.3)

Form the complex signal

$$z(t) = x(t) + j\hat{x}(t)^2 \quad (\text{Mitra 1.8})$$

where $\hat{x}(t) = x(t) \otimes h(t)$:

$$Z(j\Omega) = \begin{cases} X(j\Omega) + X(j\Omega), & \Omega \geq 0 \\ X(j\Omega) - X(j\Omega), & \Omega < 0 \end{cases} = \begin{cases} 2X(j\Omega), & \Omega \geq 0 \\ 0, & \Omega < 0 \end{cases} \quad (\text{Mitra 1.9})$$

¹Ideal Hilbert transform filtering has $|H(\Omega)| = 1$ for all Ω (**allpass** magnitude): phase $-\frac{\pi}{2}$ at each positive frequency and $+\frac{\pi}{2}$ at each negative frequency. This is why Mitra calls it a **90-degree phase shifter**. Contrast §5.1 (differentiator), where $|H| \propto |\Omega|$. Cf. J. O. Smith, "The Analytic Signal and Hilbert Transform Filters" (§5.3.4).

²The signals $x(t)$ and $\hat{x}(t)$ are called, respectively, the **in-phase** and **quadrature** components of $z(t)$.

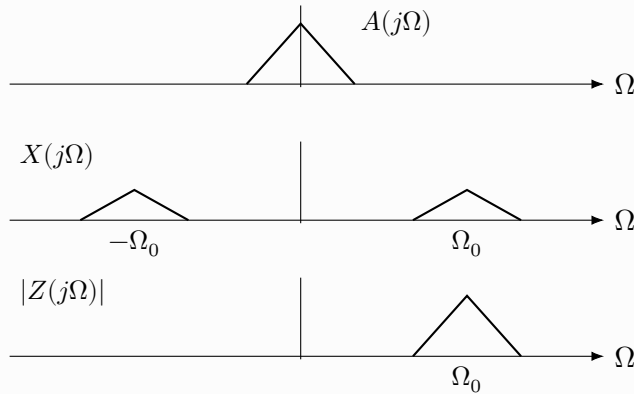
⇒ created a complex-valued signal with a **single-sided** Fourier transform (the *analytic signal*)³.

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Example (Amplitude Modulation)

(Mitra Sec. 1.2.4).

Let $a(t)$ be a lowpass signal with spectrum $A(j\Omega)$, and $x(t) = a(t) \cos(\Omega_0 t + \phi)$.



Assuming the bandwidth of $a(t)$ is smaller than Ω_0 , form the analytic signal from §5.2 ($z = x + j\hat{x}$, (1.8)):

$$z(t) = x(t) + j\hat{x}(t), \quad \hat{x}(t) = \mathcal{H}\{x(t)\}.$$

The Hilbert transformer applies $-j$ on $\Omega \geq 0$ (§5.2); for a **narrowband** real AM waveform, the envelope varies slowly compared with the carrier, so the phase shift acts on $\cos(\Omega_0 t + \phi)$ while $a(t)$ is treated as fixed:

$$\hat{x}(t) = \mathcal{H}\{a(t) \cos(\Omega_0 t + \phi)\} \approx a(t) \mathcal{H}\{\cos(\Omega_0 t + \phi)\} = a(t) \sin(\Omega_0 t + \phi).$$

Carrier check ($\theta = \Omega_0 t + \phi$): $\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$; the $+\Omega_0$ and $-\Omega_0$ terms pick up

³Why form $z = x + j\hat{x}$? The analytic signal can be written $z(t) = A(t)e^{j\psi(t)}$, with $|z(t)| = A(t)$ the **instantaneous envelope** and $\psi(t)$ the instantaneous phase (so $f(t) = \frac{1}{2\pi} \frac{d\psi}{dt}$ is instantaneous frequency). That is the usual motivation for envelope detection, AM demodulation, and related bandpass processing once $\hat{x} = \mathcal{H}\{x\}$ is available. Cf. accepted answer to “Meaning of Hilbert Transform”: <https://dsp.stackexchange.com/questions/25845/meaning-of-hilbert-transform>.

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$-j$ and $+j$ (§5.2), so $\mathcal{H}\{\cos \theta\} = \sin \theta$. With $\bullet e^{j\theta} = \cos \theta + j \sin \theta$,

$$z(t) = a(t) \cos \theta + j a(t) \sin \theta = a(t) e^{j(\Omega_0 t + \phi)}.$$

Hence

$$\Rightarrow \begin{cases} |z(t)| = |a(t)| & \text{(envelope detection),} \\ a(t) = z(t) e^{-j(\Omega_0 t + \phi)} & \text{(demodulation).} \end{cases}$$

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4 Phase Delay and Group Delay

Phase Delay.

$$\tau_p(\omega_0) \triangleq -\frac{\theta(\omega_0)}{\omega_0}. \quad (\text{Mitra 4.95})$$

Indeed, for input $e^{j\omega_0 n}$,

$$\begin{aligned} e^{j\omega_0 n} \rightarrow \boxed{h[n]} &\rightarrow |H(e^{j\omega_0})| e^{j(\omega_0 n + \theta(\omega_0))} = |H(e^{j\omega_0})| e^{j\omega_0 \left(n - \left(\frac{-\theta(\omega_0)}{\omega_0}\right)\right)} \\ &= |H(e^{j\omega_0})| e^{j\omega_0 (n - \tau_p(\omega_0))}. \end{aligned}$$

Group Delay.

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega}. \quad (\text{Mitra 4.96})$$

- **Constant group delay:** relative phases of different frequency components will be maintained.
- **GLP:** constant $\tau_g(\omega)$ for almost all ω (except at the jumps of π).
- **Non-constant phase or group delay:** distortion.⁴

⁴Example: HW4 Problem 5(b)—filters (2), (4), (6), and (7) from that set's LTI list. On a symmetric test input, $\tau_g(\omega)$ varies with ω , frequency components misalign, and the pulse shape is warped (GLP filters (1) and (5) preserve symmetry).

8 5 All-Pass Filters

All-Pass Filter.

$|H(e^{j\omega})| = 1$, while $\theta(\omega)$ is typically **not** linear. (All-pass filters can compensate nonlinear phase from other systems; that is why we sometimes intentionally add all-pass (nonlinear-phase) stages for **phase equalization**, without changing magnitude.)

Example (LDE).

$$\sum_{k=0}^M d_k y[n-k] = \sum_{k=0}^M d_{M-k}^* x[n-k].$$

If the frequency response is convergent,

$$H(e^{j\omega}) = \frac{\sum_{k=0}^M d_{M-k}^* e^{-j\omega k}}{\sum_{k=0}^M d_k e^{-j\omega k}}, \quad |H(e^{j\omega})| = 1.$$

Remark

(addendum to orig lecture note).

Why $|H| = 1$ in general: substituting $m = M - k$ in the numerator,

$$\sum_{k=0}^M d_{M-k}^* e^{-j\omega k} = e^{-j\omega M} \sum_{m=0}^M d_m^* e^{+j\omega m} = e^{-j\omega M} \left(\sum_{m=0}^M d_m e^{-j\omega m} \right)^*,$$

so the numerator and denominator have equal magnitude. The example below checks this directly.

Example.

$$d_k = \{1, \frac{1}{2}\}_{\uparrow}:$$

$$y[n] + \frac{1}{2} y[n-1] = \frac{1}{2} x[n] + x[n-1] \iff y[n] = -\frac{1}{2} y[n-1] + \frac{1}{2} x[n] + x[n-1].$$

Now we prove that this is an all-pass filter:

$$H(e^{j\omega}) = \frac{\frac{1}{2} + e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}} = \frac{\frac{1}{2} + \cos \omega - j \sin \omega}{1 + \frac{1}{2} \cos \omega - \frac{j}{2} \sin \omega},$$

We take the square of the magnitude:

$$\begin{aligned} |H(e^{j\omega})|^2 &= \frac{\left(\frac{1}{2} + \cos \omega\right)^2 + \sin^2 \omega}{\left(1 + \frac{1}{2} \cos \omega\right)^2 + \left(\frac{1}{2} \sin \omega\right)^2} \quad 5 \\ &= \frac{\frac{1}{4} + \cos \omega + \overbrace{\cos^2 \omega + \sin^2 \omega}^{=1}}{1 + \cos \omega + \frac{1}{4} \cos^2 \omega + \frac{1}{4} \sin^2 \omega} = \frac{\frac{5}{4} + \cos \omega}{\frac{5}{4} + \cos \omega} = 1. \end{aligned}$$

9

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 5.2
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 6

⁸Use two facts:

- $\left|\frac{A}{B}\right|^2 = \frac{|A|^2}{|B|^2}$
- For $z = x + jy$: $|z|^2 = x^2 + y^2$

Expanding with Euler's formula $e^{-j\omega} = \cos \omega - j \sin \omega$:

- Numerator: $\frac{1}{2} + \cos \omega - j \sin \omega \Rightarrow |\cdot|^2 = \left(\frac{1}{2} + \cos \omega\right)^2 + \sin^2 \omega$
- Denominator: $1 + \frac{1}{2} \cos \omega - \frac{j}{2} \sin \omega \Rightarrow |\cdot|^2 = \left(1 + \frac{1}{2} \cos \omega\right)^2 + \left(\frac{1}{2} \sin \omega\right)^2$
- The minus signs vanish when squared
- Dividing the two gives the first step: $|H(e^{j\omega})|^2 = \frac{\left(\frac{1}{2} + \cos \omega\right)^2 + \sin^2 \omega}{\left(1 + \frac{1}{2} \cos \omega\right)^2 + \left(\frac{1}{2} \sin \omega\right)^2}$